Indus Weave Textiles — HR Data Pipeline & Payroll Audit

PeopleOps Data Hackathon · six months of HCM exports (Apr–Sep 2026) · Python 3.14 + SQL Server 2025 Express

One command rebuilds everything: python -X utf8 pipeline/run_pipeline.py

Pipeline in one command — six stages, 13 steps

345,280 punches reconcile to the row; every raw export untouched

Stage 1 — ingest 12 files exactly as delivered · columns as text, provenance columns, hash-based idempotence

Stage 2 — seven cleaning bullets, every fix in CODE:

- 3 mixed date formats → DATE · 1,316 duplicate-key rows → 1,266 employees + 50-row merge log
- punches: 338,318 mapped → 322,143 kept / 23,137 rejected, each with a written reason
- faulty device DEV-SITE-01 (+60 min since 1 Jul) → 45,803 punch times corrected
- leave JSON flattened: 5,372 requests → 8,604 paid single days (rules 16–18)

Stage 3 — staging.attendance_daily: 217,584 employee-days, six statuses (rule 12)

Stage 4 — star schema: 4 dims (dim_employee SCD2 = 3,686 rows) + 4 facts, all reconciled

Stage 5 — 10 business questions, one .sql + one result CSV each

Stage 6 — 68 automated checks → quality.dq_results, gating the pipeline

Data quality: 68 checks, 0 blocking

raw = cleaned + rejected must hold for EVERY table — or the build fails

PK (22) — uniqueness + not-null on every key

FK (18) — every fact row finds its dimension row (both natural and surrogate keys)

Reconciliation (13) — employees 1,316 = 1,266 + 50 · punches 345,280 = 322,143 + 23,137 · JSON documents counted with OPENJSON

Accepted values (8) — statuses, leave types, pay components inside whitelist (rule 12/16/21)

Sanity (7) — no negative hours, no shift > 16 h, no overlapping SCD2 ranges

1 documented quirk = EXPECTED, not hidden: E11259 zero-length SCD2 tie (planted in source)

Planted issues fixed in code and documented: date soup · ghost duplicates · unmapped contractor punches · 2-min double punches · device clock drift · phantom 1900-01-01 effective_to sentinels

Every check WRITES its result to the DB and fails the run — quality is a gate, not a report.

Finding 1 — Stitching is paid 56% more OT than the badges show

q06.sql · paid = payroll OVERTIME units (rule 20) vs computed = punch ot_hours (rules 14–15)

D20 Stitching: paid 7,659.0 h vs computed 4,902.0 h = +56.24%

PKR 3,771,337 of OT paid to D20 — the largest OT bill in the plant (next: Weaving 1.82M)

Only department above the 15% threshold

Approvals match payment (7,656 h approved vs 7,659 paid) — so the gap is NOT the approval chain

Hypothesis to confirm by audit: pay-figure entry — the units typed into payroll

Action: audit D20's six OT months unit-by-unit against clock data; project computed OT next to paid units at every payroll run.

Finding 2 — Ghost pay: 5 employees paid after leaving

q08.sql · strictly-after-separation months (rule 4) · cross-checked by q07's zero-attendance logic

5 employees · 21 payroll months · PKR 1,948,000 paid after separation

Worst case: E10471 Umair Baloch — separated 2026-04-23, PAID May–Sep = PKR 660,000

E11222 PKR 418,100 · E10332 PKR 322,000 · E10989 PKR 294,100 · E10295 PKR 253,800

Airtight: two INDEPENDENT queries agree — separation-date comparison (q08) and attendance anti-join (q07) both find exactly 21 (employee, month) rows

Catchable at the first monthly close: PKR 132,000 (May) + PKR 454,000 (June) — ≈ PKR 0.59M, not 1.95M; the q07 query is all it takes

Action: block all 5 from next payroll; recovery letters for PKR 1,948,000; make offboarding a same-day, payroll-blocking event.

Finding 3 — 20 employees burned a full year's leave in 6 months

q10.sql · approved Annual days Apr–Sep vs FULL-YEAR entitlement (rule 18, Sundays excluded)

E10770 Danish Iqbal: 28 days used vs 14 entitlement — 2× the year in half a year

19 more employees over entitlement (E10826 27/14, E10748 25/14, ...)

Approvals exceeding entitlement means the balance check simply isn't in the loop

Deliverables: README (rebuild in 1 command) · cfo_findings.md · ERD · HTML dashboard (bonus B4)

Recommendation: enforce the balance gate in the approval UI; audit every approval that pushed past entitlement.